

Lu Yuhang

Phone number: (+86) 184-5652-5458 / (+65) 89412740 | Email: lu012933@126.com

Education Background

2024. 12 – Now	National University of Singapore	Industrial System Engineering & Management	Research Assistant
Advisor: Professor Qin Hanzhang			
Designed and implemented an LLM-based optimization framework for operations research, leveraging GPT-4.1 architecture with domain-specific instruction sets. Developed a multi-agent RAG pipeline integrating FAISS vector search retriever and LangChain tools to process multimodal data (including 10k+ row user-uploaded tables), enabling automatic coefficient extraction and large-scale modeling. Eliminated the need for costly model fine-tuning through dynamic few-shot learning and chain-of-thought prompting. Delivered solver-ready LP models, executable Gurobi Python code, and optimal solutions for logistics optimization.			
2023. 08 – 2024.12	National University of Singapore	Industrial System Engineering & Management	Master
Advisor: Professor Qin Hanzhang			
Main Courses: Applied Programming for Industrial Systems: A Stochastic Models and Their Applications: A Modern Supply Chain and Blockchain Technologies: A Data Analytics for Industrial Engineers: A- Applied Engineering Statistics: A			
2019. 08 - 2023. 06	Xi'an Jiaotong University	Science and Technology of Human Settlements (Big Data & Smart City)	Bachelor
Average Score: 87.59/100 GPA: 3.65/4.0 Rank: 2/22			
Main Courses: Internet of Things Application Technology 97/100, Urban Sociology 97.5/100, Collaborative Urban Design for Built Environment in Big Data 94/100, Big Data and Data Mining Techniques for Engineering Application 92/100, Electrotechnics and Electronics 97/100			
Awards: Second grade scholarship (Top 10% in major), Outstanding Student, Outstanding Graduate Student			
2021. 08 - 2021. 12	University of California, Berkeley	Urban Planning and Environment Data Analysis	Exchange
Main Course: Geospatial Technologies: Mapping, Space and Power (A+), Programming in R(A-), Applied Time Series Analysis for Ecology and Environmental Sciences (B+)			

Research Experience

Working Paper: Multi-Agent LLM Framework for Reasoning and Behavioral Benchmark Evaluation	Jan 2026 – Present
➤ Independently designed and implemented a Python-based multi-agent LLM experimentation framework to compare five collaboration paradigms, including self-reflection, debate, reconciliation, role-specialized reasoning, and chain-of-agents, on GSM8K, CRT, and Linda benchmarks. ➤ Built a unified model interface for Ollama/OpenAI-style backends, task-specific prompt templates and answer parsers, plus Excel-based summary/trace export and a Streamlit dashboard for batch orchestration, run-plan visualization, and result inspection. ➤ Executed 119 experiment runs across heterogeneous model combinations, analyzed accuracy and runtime trade-offs, and created reproducible evaluation artifacts with automated tests for experiment scheduling logic.	
Working Paper: LEAN-LLM-OPT: A Lightweight Multi-Agent LLM Framework for Large-Scale Automated Optimization Modeling Aug 2023 – Present	
SSRN link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5329027	
➤ Independently designed a multi-agent LLM-based automated optimization modeling framework for solving large-scale optimization problems (e.g., network flow, path planning, knapsack problems). Based on GPT-4.1, integrated domain-specific instruction sets for classical OR models, significantly enhancing the model's natural language understanding and generation capabilities for linear programming (LP) scenarios, resulting in high-fidelity outputs. ➤ Integrated Kaggle datasets as reference corpora. Implemented a Multi-Agent Retrieval-Augmented Generation (RAG) pipeline using FAISS for vector similarity search, and custom RAG tools. Employed a RAG agent for initial user query classification. Leveraged retrieved similar case embeddings within prompts for few-shot learning. Retrieved and processed user-uploaded external data via RAG, utilizing valid data to formulate optimization models. Output standardized optimization models (LP format), executable Gurobi Python code, and optimal solutions. Monitored test set accuracy throughout the process using LangSmith and OpenAI API.	
➤ Achieved Key Breakthroughs: ➤ Large-Scale Modeling Capability: Enabled processing of external large-scale tabular data files (10k+ rows) via RAG retrieval. The model performs precise table information retrieval, extracts relevant user-concerned field data as coefficients, performs calculations, and formulates complex optimization algorithms.	

- **Parameter-Efficient, Low-Cost Training:** Achieved robust performance using minimal examples combined with RAG similarity retrieval to automatically construct optimal few-shot learning prompts. Leveraged prompt engineering and Chain-of-Thought (CoT) multi-agent structure reasoning to guide the model's problem-solving process, eliminating the need for costly and time-consuming fine-tuning.
- **Generalizability & Usability:** The model outputs optimization models in standard LP format compatible with solvers and supports Markdown/LaTeX rendering. It subsequently generates corresponding Gurobi Python code. Capable of handling complex optimization scenarios in logistics and related domains.
- Executed data preprocessing and instruction set generation to optimize model training quality. Achieved 95.1% accuracy on the NL4OPT benchmark dataset. Demonstrated competitive performance against baseline models (GPT-4o, ORLM) on the IndustryOR industrial dataset.

Research Project: Multivariable Deep Learning Model for European Natural Gas Forecasting

Feb 2024 – Jun 2024

- European natural gas market analysis model: Integrated various data, including economic indices, stock market volatility, weather changes, alternative energy usage, and geopolitical factors, with Title Transfer Facility (TTF) price and market fundamentals analysis (supply-demand balance).
- Applied deep learning frameworks—LSTM, Random Forest, and the innovative PatchTSMixer model—for accurate prediction of multi-dimensional time series. Research variables included European stock prices, natural gas import/export volumes, storage levels, natural gas demand (industrial, residential, power generation), demand and price of alternative energy, weather temperature data, and Euro exchange rate.
- Fine-tuned the Llama3-8b model, with an expanded corpus focusing on natural gas-specific language. Analyzed all political factors related to natural gas prices since the inception of the European natural gas market, modified sentiment analysis algorithms to calculate the impact weight of different phrases on prices, and fine-tuned the model with text instructions related to the news impact on price fluctuations.
- Developed an HTML visualization page and an open-web UI for user-friendly interaction for local deployment, effectively enhancing the accuracy of TTF futures price forecasting and providing data-driven insights for strategic decisions in the natural gas market.

Internship

2020. 07 – 2020. 09

iFlyTek

Assistant Account Manager

China, Hefei

- Assist customer manager to complete customer service management; Participate in helping to establish and maintain good business relationship with customers, feedback customer use of the company's products related information.
- Participate in team projects, assist colleagues to complete the research of iFlyTek's organizational structure, development model, and existing business group management and control mode. Participated in the survey and sympathy of outsourcing companies, and participated in the internal OA process, personnel management, supply chain services and other aspects of the company.
- Maintained and updated iFlyTek personnel database and public account, participated in performance meetings and personnel management meetings for many times, assisted in meeting layout and equipment retrieval, and wrote meeting minutes and comment reports.

Competition

2021. 07 – 2022. 06

National project on Undergraduate Innovation and Entrepreneurship

National Excellent * 3

- Won national excellent prize in the following three projects: *“I love Chemistry” Applet: Spread Chemical Culture, Cultivate Students’ Ability; “Sand Fixation”——Design of intelligent auxiliary desert management system”*; *“shuizhuchenfu——A self-actuated bionic underwater energy harvesting power supply device with thermoelectric-piezoelectric hierarchical coupling is presented”*
- Lead the team to complete product feature analysis, work environment tolerance analysis, task index completion analysis.
- Participate in the functional logic design, coding and UI of the public account, independently complete dubbing, editing, rendering and other work, and launch the public account. Integrated university chemistry teaching materials and communicated with professors about experimental shooting layout and scientific verification.
- Complete project planning and construction. Code the control and automation parts of the system. Skilled use of Final Cut Pro, film cutting and other related tools to complete video editing, dubbing, subtitling, rendering and other work.

National University Student Social Practice and Science Contest on Energy Saving & Emission Reduction

2021.09 – 2022.08

Experiment operation, Data processing, Visual presentation, Video Editing

National First Prize

- Responsible for the construction of large-scale experimental platform, equipment debugging and maintenance, and managed the laboratory to complete scheduled water supply and drainage. Using GraphPad Prism 9 and Excel to process visual data. Using C4D, SOLIDWORKS for scene modeling and animation making.
- Completed video shooting independently, skillfully used Final Cut Pro to complete video editing, rendering, production and other work.
- Successfully completed the contest defense, stood out from the works of 1500+ participating teams, and won the national first prize (the highest award).
- Declared a patent for an invention related to environmental energy, patent No. 202210547923.3, which is applicable to capture various types of ocean current energy

in the deep ocean through piezoelectric and thermoelectric devices, and complete self-energy for deep-sea exploration such as sonar.

Club and Organization Experience

2021.10—Now	Global Green Development Alliance (GLOGDA)	Youth group member	California, US
--------------------	---	---------------------------	-----------------------

- Participated in the joint project of the State Grid Corporation of China in the California Department of the United States, assisted participating members discussing the status quo of energy transformation, and studied the cutting-edge technology and equipment of new energy science and technology.
- Organize members to hold alliance meetings, coordinate and communicate with the Youth Department, Youth Group, Adult Committee and other departments, and connect professors from many well-known universities, including Stanford and Berkeley, California, to participate and hold lectures.
- Work with members to establish the business composition, team nature and work direction of the Global Green Development Alliance to provide a platform for global leaders, professional bodies, governments and industry to integrate policy, business, investment and research and development projects.
- Assisted in the holding of Youth Leadership Summit, completed issuance of conference member invitations, summit poster production, Bilibili live broadcast, YouTube live broadcast, public account writing and publishing, website management and other work. Awarded the title of outstanding volunteer.